

Project Executable Files

1. Introduction

The Smart Lender – Loan Eligibility Prediction System is a Machine Learning-based web application developed using Python and Flask. The project includes executable files, trained machine learning models, datasets, and supporting resources required to run the application successfully.

This document describes the files necessary to execute the Smart Lender application.

2. Project Files

File / Folder	Purpose
app.py	Main Flask application used to run the project
requirements.txt	Contains all required Python libraries
model/saved_model.pkl	Trained Random Forest machine learning model
model/scaler.pkl	Feature scaling model used during prediction
templates/	HTML pages for the user interface
Dataset/	Loan prediction dataset used for training
Training/	Model training scripts and EDA outputs

3. Software Requirements

Software	Version
Python	3.x
Visual Studio Code	Latest
Flask	Latest
Git	Latest
Google Chrome	Latest

4. Required Python Libraries

- Flask
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Imbalanced-learn (SMOTE)
- Pickle

These dependencies are listed in the requirements.txt file and can be installed using:

```
pip install -r requirements.txt
```

5. Steps to Execute the Project

Step 1

Clone the repository from GitHub.

Step 2

Open the project folder in Visual Studio Code.

Step 3

Install all required dependencies:

```
pip install -r requirements.txt
```

Step 4

Run the Flask application:

```
python app.py
```

Step 5

Open the application in your web browser using the local Flask URL.

Step 6

Enter applicant details and click **Predict** to view the loan eligibility result.

6. Expected Output

The application processes the user input through the trained Random Forest model and displays one of the following outcomes:

-  Loan Approved
 -  Loan Rejected
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7. Project Deliverables

- Flask Web Application
 - Trained Machine Learning Model
 - Loan Prediction Dataset
 - Source Code
 - Project Documentation
 - GitHub Repository
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8. Conclusion

The Smart Lender project includes all necessary executable files and supporting resources required to run the application successfully. By following the setup and execution steps, users can easily launch the system and perform loan eligibility predictions through the web interface.